**1) a)**

$(2n+1)(2n-3)-4n+23 = 4n^2-4n-3-4n+23$

$= 4n^2-8n+20$

$= 4(n^2-2n+5)$

$\therefore (2n+1)(2n-3)-4n+23$ is divisible by $4$

**1) b)**

$16^{3k}-4 = (4^2)^{3k}-4 = 4^{6k}-4 = 4(4^{6k-1}-1)$

$\therefore 16^{3k}-4$ is divisible by $4$

**1) c)**

$2^{k+2}-4\times 3^k+8 = 2^2\times 2^k-4\times 3^k+8 = 4(2^k-3^k+2)$

$\therefore 2^{k+2}-4\times 3^k+8$ is divisible by $4$

**2) a)**

$6^{k+1}-18 = 6\times 6^k-18 = 6(6^k-3)$

$\therefore 6^{k+1}-18$ is divisible by $6$

**2) b)**

$2^{k+1}\times 3^{k+2} = 2\times 2^k\times 3\times 3^{k+1} = 6(2^k\times 3^{k+1})$

$\therefore 2^{k+1}\times 3^{k+2}$ is divisible by $6$

**2) c)**

$(k^2+1)(k-4)+10(k^2+k+1) = k^3-4k^2+k-4+10k^2+10k+10$

$= k^3+6k^2+11k+6$

$= (k+1)(k+2)(k+3)$

which is $3$ consecutive integers.

At least one of these factors will be even so the expression is a multiple of $2$.

One of the factors will be a multiple of $3$.

As the expression is a multiple of both $2$ and $3$, it must be a multiple of $6$.

$\therefore (k^2+1)(k-4)+10(k^2+k+1)$ is divisible by $6$

**3) a)**

False: if $a = 2$, $b = 3$ and $c = 6$ then $a + b = 5$ which does not divide $6$

**3) b)**

True:

Let $pa = b$ and $qa = c$ for some values $p$ and $q$

Then $b + c = pa + qa = a(p + q)$ so $a$ is a factor of $b + c$

**3) c)**

False: if $a = 5$, $b = 2$ and $c = 3$ then $b + c = 5$ so $a$ is a factor of $b + c$ but $b$ and $c$ are not factors of $a$

**3) d)**

True:

Let $pa = b$ and $qb = c$ for some values $p$ and $q$

Then $b + c = pa + qb = pa + pqa = a(p + pq)$ so $a$ is a factor of $b + c$

**4)**

Prove true for $n = 1$:

$1 + 1 = 12$ which is divisible by 2 so true for $n = 1$

Assume true for $n = k$:

$k^2 + k$ is divisible by 2

Prove true for $n = k + 1$:

$(k + 1)^2 + (k + 1)$

$= k^2 + 2k + 1 + k + 1$

$= k^2 + 3k + 2$

$= (k^2 + k) + 2k + 2$

$= (k^2 + k) + 2(k + 1)$

Since we assumed $k^2 + k$ is divisible by 2 and $2(k + 1)$ is divisible by 2

$(k^2 + k) + 2(k + 1)$ must be divisible by 2

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**5) a)**

Prove true for $n = 1$:

$1 - 1 = 0$ which is divisible by 3 so true for $n = 1$

Assume true for $n = k$:

$k^3 - k$ is divisible by 3

Prove true for $n = k + 1$:

$(k+1)^3 - (k+1)$

$= k^3 + 3k^2 + 3k + 1 - k - 1$

$= k^3 + 3k^2 + 2k$

$= (k^3 - k) + 3k^2 + 3k$

$= (k^3 - k) + 3(k^2 + k)$

Since we assumed $k^3 - k$ is divisible by 3 and $3(k^2 + k)$ is divisible by 3

$(k^3 - k) + 3(k^2 + k)$ must be divisible by 3

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**5) b)**

Prove true for $n = 1$:

$1 + 3 + 2 = 6$ which is divisible by 6 so true for $n = 1$

Assume true for $n = k$:

$k^3 + 3k^2 + 2k$ is divisible by 6

Prove true for $n = k + 1$:

$(k+1)^3 + 3(k+1)^2 + 2(k+1)$

$= k^3 + 3k^2 + 3k + 1 + 3k^2 + 6k + 3 + 2k + 2$

$= k^3 + 6k^2 + 11k + 6$

$= (k^3 + 3k^2 + 2k) + 3k^2 + 9k + 6$

$= (k^3 - k) + 3(k^2 + 3k + 2)$

$= (k^3 - k) + 3(k+1)(k+2)$

Since we assumed $k^3 - k$ is divisible by 6, $(k + 1)$ and $(k + 2)$ are consecutive integers so one must be even so that $3(k+1)(k+2)$ must be divisible by both 2 and 3 therefore divisible by 6

$(k^3 - k) + 3(k+1)(k+2)$ must be divisible by 6

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**6)**

Prove true for $n = 1$:

$9^1 + 7 = 16$ which is divisible by $8$ so true for $n = 1$

Assume true for $n = k$:

$9^k + 7$ is divisible by $8$

Prove true for $n = k + 1$:

$9^{k+1} + 7$

$= 9 \times 9^k + 63 - 56$

$= 9(9^k + 7) - 56$

Since we assumed $9^k + 7$ is divisible by $8$ and $56$ is divisible by $8$ then $9(9^k + 7) - 56$ must be divisible by $8$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**7)**

Prove true for $n = 1$:

$5^1 - 4 + 3 = 4$ which is divisible by $4$ so true for $n = 1$

Assume true for $n = k$:

$5^k - 4k + 3$ is divisible by $4$

Prove true for $n = k + 1$:

$5^{k+1} - 4(k + 1) + 3$

$= 5 \times 5^k - 4k - 1$

$= 5 \times 5^k - 20k + 15 + 16k - 16$

$= 5(5^k - 4k + 3) + 4(4k - 4)$

Since we assumed $5^k - 4k + 3$ is divisible by $4$ and $4(4k - 4)$ is divisible by $4$ then $5(5^k - 4k + 3) + 4(4k - 4)$ must be divisible by $4$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**8)**

Prove true for $n = 1$:

$5^1 + 4 - 1 = 8$ which is divisible by 8 so true for $n = 1$

Assume true for $n = k$:

$5^k + 4k - 1$ is divisible by 8

Prove true for $n = k + 1$:

$5^{k+1} + 4(k + 1) - 1$

$= 5 \times 5^k + 4k + 3$

$= 5 \times 5^k + 20k - 5 - 16k + 8$

$= 5(5^k + 4k - 1) - 8(2k - 1)$

Since we assumed $5^k + 4k - 1$ is divisible by 8 and $8(2k - 1)$ is divisible by 8 then $5(5^k + 4k - 1) - 8(2k - 1)$ must be divisible by 8

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**9)**

Prove true for $n = 1$:

$6^1 - 2^1 = 4$ which is a multiple of 4 so true for $n = 1$

Assume true for $n = k$:

$6^k - 2^k$ is a multiple of 4

Prove true for $n = k + 1$:

$6^{k+1} - 2^{k+1}$

$= 6 \times 6^k - 2 \times 2^k$

$= 2 \times 6^k + 4 \times 6^k - 2 \times 2^k$

$= 2(6^k - 2^k) + 4 \times 6^k$

Since we assumed $6^k - 2^k$ is a multiple of 4 and $4 \times 6^k$ is a multiple of 4 then $2(6^k - 2^k) + 4 \times 6^k$ must be a multiple of 4

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**10) a)**

Prove true for $n = 1$:

$5^1 + 3 \times 9^1 = 32$ which is divisible by $4$ so true for $n = 1$

Assume true for $n = k$:

$5^k + 3 \times 9^k$ is divisible by $4$

Prove true for $n = k + 1$:

$5^{k+1} + 3 \times 9^{k+1}$

$= 5 \times 5^k + 3 \times 9 \times 9^k$

$= 5 \times 5^k + 27 \times 9^k$

$= 5 \times 5^k + 15 \times 9^k + 12 \times 9^k$

$= 5(5^k + 3 \times 9^k) + 12 \times 9^k$

Since we assumed $5^k + 3 \times 9^k$ is divisible by $4$ and $12 \times 9^k$ is divisible by $4$ then $5(5^k + 3 \times 9^k) + 12 \times 9^k$ must be divisible by $4$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**10) b)**

Prove true for $n = 1$:

$5^2 + 3^{2-1} = 28$ which is divisible by $4$ so true for $n = 1$

Assume true for $n = k$:

$5^{2k} + 3^{2k-1}$ is divisible by $4$

Prove true for $n = k + 1$:

$5^{2(k+1)} + 3^{2(k+1)-1}$

$= 5^{2k+2} + 3^{2k+1}$

$= 25 \times 5^{2k} + 3^2 \times 3^{2k-1}$

$= 25 \times 5^{2k} + 9 \times 3^{2k-1}$

$= 25 \times 5^{2k} + 25 \times 3^{2k-1} - 16 \times 3^{2k-1}$

$= 25(5^{2k} + 3^{2k-1}) - 16 \times 3^{2k-1}$

Since we assumed $5^{2k} + 3^{2k-1}$ is divisible by $4$ and $16 \times 3^{2k-1}$ is divisible by $4$ then $25(5^{2k} + 3^{2k-1}) - 16 \times 3^{2k-1}$ must be divisible by $4$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**11)**

Prove true for $n = 2$:

$15^2 - 8^0 = 224$ which is divisible by 7 so true for $n = 2$

Assume true for $n = k$:

$15^k - 8^{k-2}$ is divisible by 7

Prove true for $n = k + 1$:

$15^{k+1} - 8^{k+1-2} = 15 \times 15^{k+1} - 8 \times 8^{k-2}$

$= 15 \times 15^k - 15 \times 8^{k-2} + 7 \times 8^{k-2}$

$= 15(15^k - 8^{k-2}) + 7 \times 8^{k-2}$

Since we assumed $15^k - 8^{k-2}$ is divisible by 7 and $7 \times 8^{k-2}$ is divisible by 7 then $15(15^k - 8^{k-2}) + 7 \times 8^{k-2}$ must be divisible by 7

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**12)**

Prove true for $n = 1$:

$3^1 + 7^1 = 10$ which is divisible by 10 so true for $n = 1$

Assume true for $n = k$ where $k$ is odd

$3^k + 7^k$ is divisible by 10

Prove true for $n = k + 2$:

$3^{k+2} + 7^{k+2} = 3^2 \times 3^k + 7^2 \times 7^k$

$= 9(3^k + 7^k) + 40 \times 7^k$

Since we assumed $3^k + 7^k$ is divisible by 10 and $40 \times 7^k$ is divisible by 10 then $9(3^k + 7^k) + 40 \times 7^k$ must be divisible by 10

As the statement is true for $n = 1$, and if the statement is true for $n = k$ where $k$ is odd, then the statement is true for $n = k + 2$, so the statement is true for all odd integers $n \geq 1$

**13) a)**

Errors:

Didn't check true for $n = 1$

Didn't state "assume true for $n = k$"

Only attempted to show that the difference in two terms is divisible by $8$ rather than specifically looking at the $k + 1$ case.

**13) b)**

Let $\mathrm{f}(k) = 3^{2k} - 1$

Prove true for $n = 1$:

$3^2 - 1 = 8$ which is a multiple of $8$ so true for $n = 1$

Assume true for $n = k$:

$3^{2k} - 1$ is a multiple of $8$

Prove true for $n = k + 1$:

$\mathrm{f}(k + 1) - \mathrm{f}(k) = \left(3^{2(k+1)} - 1\right) - (3^{2k} - 1)$

$\mathrm{f}(k + 1) - 8m = 3^2 \times 3^{2k} - 3^{2k}$ where $m$ is some integer

$\mathrm{f}(k + 1) = 8m + (9 - 1) \times 3^{2k}$

$\mathrm{f}(k + 1) = 8m + 8 \times 3^{2k}$

$\mathrm{f}(k + 1) = 8[m + 3^{2k}]$

which is divisible by $8$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**14)**

Let three consecutive positive integers be $n$, $n+1$ and $n+2$ therefore proving $n^3+(n+1)^3+(n+2)^3$ is divisible by 3

Prove true for $n=1$:

$1^3+2^3+3^3=36$ which is divisible by 3 so true for $n=1$

Assume true for $n=k$

$k^3+(k+1)^3+(k+2)^3$ is divisible by 3

Prove true for $n=k+1$:

$(k+1)^3+(k+2)^3+(k+3)^3$

$=k^3+3k^2+3k+1+k^3+6k^2+12k+8+k^3+9k^2+27k+27$

$=3k^3+18k^2+42k+36$

$=3[k^3+6k^2+14k+12]$

which is divisible by 3

As the statement is true for $n=1$, and if the statement is true for $n=k$ then the statement is true for $n=k+1$, so the statement is true for all integers $n\geq 1$

**15) a)**

$f(n+1) - f(n) = 50(n+1)^2 + 70(n+1) - (50n^2 + 70n)$

$= 50n^2 + 100n + 50 + 70n + 70 - 50n^2 - 70n$

$= 100n + 120$

$= 20(5n + 6)$ as req.

**15) b)**

Prove true for $n = 1$:

$f(1) = 50 + 70 = 120$ which is divisible by $4$ so true for $n = 1$

Assume true for $n = k$:

so let $f(k) = 4m$ for some integer $m$

Prove true for $n = k + 1$:

$f(k+1) - f(k) = 20(5k + 6)$

$f(k+1) - 4m = 20(5k + 6)$

$f(k+1) = 4m + 20(5k + 6)$

$f(k+1) = 4[m + 5(5k + 6)]$

which is divisible by $4$

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**16) a)**

$f(n+1) - f(n) = 24 \times 2^{4(n+1)} + 3^{4(n+1)} - 24 \times 2^{4n} - 3^{4n}$

$= 24 \times 2^{4n+4} + 3^{4n+4} - 24 \times 2^{4n} - 3^{4n}$

$= 384 \times 2^{4n} + 81 \times 3^{4n} - 24 \times 2^{4n} - 3^{4n}$

$= 360 \times 2^{4n} + 80 \times 3^{4n}$

**16) b)**

Prove true for $n = 1$:

$f(1) = 24 \times 2^4 + 3^4 = 465$ which is divisible by 5 so true for $n = 1$

Assume true for $n = k$:

so let $f(k) = 5m$ for some integer $m$

Prove true for $n = k + 1$:

$f(k+1) - f(k) = 360 \times 2^{4k} + 80 \times 3^{4k}$

$f(k+1) - 5m = 360 \times 2^{4k} + 80 \times 3^{4k}$

$f(k+1) = 5m + 360 \times 2^{4k} + 80 \times 3^{4k}$

$f(k+1) = 5[m + 72 \times 2^{4k} + 16 \times 3^{4k}]$

which is divisible by 5

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**17) a)**

Prove true for $n = 1$:

$2 - 5 = -3$ which is a multiple of 3 so true for $n = 1$

Assume true for $n = k$:

$2k^3 - 5k$ is a multiple of 3

Prove true for $n = k + 1$:

$2(k+1)^3 - 5(k+1)$

$= 2k^3 + 6k^2 + 6k + 2 - 5k - 5$

$= 2k^3 + 6k^2 + k - 3$

$= (2k^3 - 5k) + 6k^2 + 6k - 3$

$= (2k^3 - 5k) + 3(2k^2 + 2k - 1)$

Since we assumed $2k^3 - 5k$ is a multiple of 3 and $3(2k^2 + 2k - 1)$ is a multiple of 3 $(2k^3 - 5k) + 3(2k^2 + 2k - 1)$ must be a multiple of 3

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$

**17) b)**

$f(-x) = 2(-x)^3 - 5(-x) = -2x^3 + 5x = -(2x^3 - 5x) = -f(x)$

**17) c)**

From part **a)**, $f(n)$ is a multiple of 3 for all integers $n \geq 1$

From part **b)**, $f(-n) = -f(n)$ so is a multiple of 3 so is true for all integers $n \leq 1$

$f(0) = 0$ which is a multiple of 3

$\therefore f(n)$ is a multiple of 3 for all integer values of $n$